

Cybersecurity Bayesian Network: Phishing Detection

Kyana Marckx (852793982) - Bayesian Reasoning & Learning (IM0902)

July 12, 2026

Contents

1 Task 1 - Build a Bayesian Network in pyAgrum	1
1.1 Introduction	1
1.2 Methods	2
1.2.1 Variable Selection	2
1.2.2 Causal Structure	2
1.2.3 Network Structure Representation	2
1.2.4 Parameter Specification	3
1.3 Implementation	3
1.4 Validation and Analysis	3
2 Task 2 - Learn a Bayesian Network in pyAgrum	4
2.1 Introduction	4
2.2 Methods	4
2.2.1 Data generation	4
2.2.2 Structure learning methods	4
2.2.3 Evaluation of structure learning	4
2.2.4 Classification setup	5
2.3 Results	5
2.3.1 Structure learning results	5
2.3.2 Classification results	5
2.4 Conclusion	5

1 Task 1 - Build a Bayesian Network in pyAgrum

1.1 Introduction

Phishing remains one of the most prevalent forms of cybercrime and continues to pose a significant threat to both individuals and organizations. Attackers typically disguise fraudulent emails as legitimate communications in an attempt to deceive users into revealing sensitive information, installing malware, or performing actions that compromise security. Despite advances in technical detection methods, phishing attacks remain highly successful because they exploit human behavior through social engineering techniques rather than purely technical vulnerabilities [AHNK21, JLN25].

Several studies have shown that phishing emails often contain identifiable indicators, such as suspicious sender information, misleading hyperlinks, poor, grammar, and language designed to create urgency or invoke authority [PBP⁺16, Nai15, Ros25]. At the same time, user-related factors such as cybersecurity awareness strongly influence whether a phishing attempt ultimately succeeds [ZBDR⁺24].

Because phishing detection involves uncertainty and incomplete information, Bayesian Networks provide a suitable modelling framework. They allow probabilistic reasoning based on observable evidence while explicitly representing causal relationships between variables. Rather than making a deterministic decision, a Bayesian Network can estimate the likelihood that an email is malicious and how different pieces of evidence contribute to that assessment.

The objective of this work is to construct a Bayesian Network capable of estimating the probability that an email is a phishing attempt and the probability that it ultimately leads to account compromise. The model combines observable characteristics of an email with behavioral characteristics of the user, enabling both prediction and explanation of phishing-related risks.

1.2 Methods

1.2.1 Variable Selection

The variables included in the network were selected based on commonly reported phishing indicators in cybersecurity literature. Research has identified sender legitimacy, hyperlinks, grammatical quality, and urgency cues as important characteristics that users rely on when distinguishing phishing emails from legitimate communications [PBP⁺16, Nai15, Ros25].

The Bayesian Network consist of eight binary variables, as shown in Table 1. All variables were modelled as binary variables in order to reduce model complexity and facilitate interpretation of the resulting probability distributions.

Variable	Description	Values
<i>SenderReputation</i>	Indicates whether the sender appears trustworthy or suspicious.	good / suspicious
<i>EmailGrammar</i>	Represents the quality of grammar and spelling used in the email.	good / poor
<i>ContainsLinks</i>	Indicates whether the email contains hyperlinks.	yes / no
<i>UrgencyLanguage</i>	Indicates whether the email contains urgent or threatening language.	yes / no
<i>PhishingEmail</i>	Indicates whether the email is a phishing attempt.	yes / no
<i>UserAwareness</i>	Represents the cybersecurity awareness level of the user	high / low
<i>ClickLink</i>	Indicates whether the user clicks a link contained in the email.	yes / no
<i>AccountCompromised</i>	Indicates whether the user’s account becomes compromised.	yes / no

Table 1: Description and values of the selected variables in the Bayesian Network

1.2.2 Causal Structure

The network structure was manually designed according to causal assumptions derived from the phishing process.

SenderReputation, *EmailGrammar*, *ContainsLinks*, and *UrgencyLanguage* were modelled as direct causes of the variable *PhishingEmail*. This reflects the assumption that these observable characteristics provide evidence regarding whether an email is malicious. Previous studies have shown that phishing attacks frequently rely on suspicious sender identities, embedded links, and urgency-based persuasion strategies to increase user compliance [JLN25, Nai15, Ros25].

The variable *ClickLink* depends on both *PhishingEmail* and *UserAwareness*. This relationship captures the idea that phishing emails are more likely to trigger user interaction, while users with higher cybersecurity awareness are generally better able to recognize and avoid phishing attempts [ZBDR⁺24].

Finally, *AccountCompromised* was modelled as a direct consequence of *ClickLink*. This reflects a simplified attack chain in which account compromise becomes substantially more likely after interaction with a malicious link.

The resulting Bayesian Network models the progression from email characteristics to phishing detection, user behavior, and ultimately account compromise.

1.2.3 Network Structure Representation

The final Bayesian Network consists of eight nodes and seven directed edges. A graphical representation of the network is shown in Figure 1.

The network follows a simplified phishing attack chain. Email characteristics influence the probability that an email is a phishing attempt. The phishing status of the email, together with the cybersecurity awareness of the user, influences the probability that a link is clicked. Finally, clicking on a malicious link increases the probability of account compromise.

This structure was intentionally kept relatively compact in order to maintain interpretability while still capturing the most important causal relationships involved in phishing attacks.

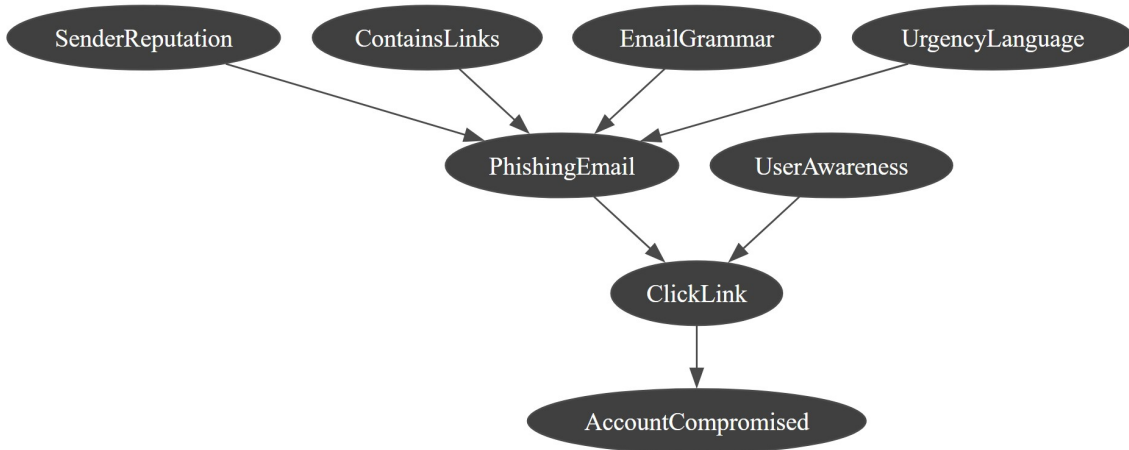


Figure 1: Bayesian Network structure for phishing detection and risk assessment

1.2.4 Parameter Specification

After defining the network structure, probability distributions were assigned to all variables. Since no complete dataset describing the selected domain was available, the parameters were specified manually using expert knowledge and observations reported in cybersecurity literature.

Prior probabilities were assigned to all root nodes. These probabilities reflect general assumptions about the prevalence of suspicious senders, poor grammar, embedded hyperlinks, urgency-based language, and cybersecurity awareness within a typical email environment.

For the variable *PhishingEmail*, a conditional probability table (CPT) was constructed based on the assumption that suspicious senders, poor grammar, embedded hyperlinks, and urgency cues all increase the likelihood of an email being a phishing attempt. Combinations containing multiple phishing indicators were therefore assigned substantially higher probabilities than combinations containing only legitimate characteristics.

The CPT of *ClickLink* was designed to reflect user susceptibility. Users with low cybersecurity awareness were assumed to be more likely to interact with links, particularly when confronted with phishing emails. Conversely, users with high awareness were assumed to be more resistant to such attacks.

Finally, the CPT of *AccountCompromised* reflects a simplified attack process in which account compromise becomes significantly more likely after a user clicks on a malicious link.

The complete CPTs used in the model are documented in the project repository¹ and were subsequently implemented in pyAgrum² for inference and data generation (see Task 2).

1.3 Implementation

The Bayesian Network was implemented using the Python library pyAgrum. The model structure was first defined by explicitly adding all variables as nodes, followed by the specification of directed edges representing causal relationships between variables.

After defining the network structure, conditional probability tables (CPTs) were manually assigned based on the expert-driven design described in the previous sections. These CPTs were encoded directly in pyAgrum using discrete probability distributions for each node.

The implemented model was then used for probabilistic inference using the LazyPropagation algorithm provided by pyAgrum. This allowed the computation of posterior probabilities under different evidence scenarios, which were used to validate the behavior of the network.

Finally, the model was exported to a .bif file format to ensure reproducibility and to enable its use in subsequent structure learning experiments in Task 2.

1.4 Validation and Analysis

The Bayesian Network was validated through a series of inference experiments designed to verify whether the resulting posterior probabilities were consistent with expected phishing behavior.

¹<https://github.com/kyanamarckx/bayesian-network/tree/main>

²<https://pyagrum.gitlab.io>

First, the prior probability of *PhishingEmail* was computed without introducing any evidence. The network estimated a baseline phishing probability of 19.31%. This value serves as a reference point for evaluating the influence of additional evidence.

To assess the impact of individual phishing indicators, the sender reputation was set to *suspicious*. Under this condition, the posterior probability of *PhishingEmail* increased from 19.31% to 50.92%. This substantial increase confirms that the network correctly interprets suspicious senders as an important indicator of phishing activity.

A second experiment combined all major phishing indicators considered in the model: suspicious sender reputation, poor grammar, embedded hyperlinks, and urgency-based language. When all four indicators were observed simultaneously, the posterior probability of *PhishingEmail* increased to 95%. This result demonstrates that the network successfully aggregates multiple sources of evidence and strongly favors the phishing hypothesis when numerous warning signs are present.

The effect of cybersecurity awareness was evaluated by comparing the probability that a user clicks a link contained in a phishing email. For users with high cybersecurity awareness, the probability of clicking a phishing link was 60%. In contrast, users with low cybersecurity awareness exhibited a click probability of 90%. These results reflect the intended modelling assumption that awareness reduces susceptibility to phishing attacks and confirms that the human factor is appropriately represented in the network.

Finally, an end-to-end attack scenario was analyzed by combining all phishing indicators with a user exhibiting low cybersecurity awareness. Under these conditions, the probability of account compromise reached 61.03%. This result illustrates how risk propagates through the causal chain represented by the Bayesian Network. Although the phishing probability of account compromise remains lower because successful compromise additionally depends on user interaction and the conditional probability of compromise after clicking a malicious link.

Overall, the inference results are consistent with the intended semantics of the model and no unexpected probabilistic behavior was observed. The experiments provide evidence that the Bayesian Network offers a plausible representation of phishing attacks and their potential consequences.

2 Task 2 - Learn a Bayesian Network in pyAgrum

2.1 Introduction

In this task, structure learning algorithms for Bayesian Networks are investigated using synthetic data generated from the original phishing detection model. The goal is to evaluate how well different learning approaches can recover the underlying network structure and how this affects classification performance.

In addition, the learned networks are compared with the original manually designed model and a Naive Bayes classifier in a predictive setting.

2.2 Methods

2.2.1 Data generation

Synthetic datasets were generated by sampling from the original Bayesian Network using pyAgrum. Three datasets sizes were used (100, 500, and 1000 samples) to study the effect of sample size on structure recovery. A separate training and test split (100 samples each) was used for classification experiments.

2.2.2 Structure learning methods

Two structure learning approaches were evaluated:

- **Search-and-score learning:** A greedy hill climbing algorithm using the Bayesian Information Criterion (BIC) as scoring function.
- **Constraint-based learning:** A dependency-based approach using mutual information (MIIC) to infer conditional dependencies between variables.

2.2.3 Evaluation of structure learning

The learned network structures were compared to the original Bayesian Network by analyzing the number of correctly recovered edges, missing edges, and additional incorrect edges.

2.2.4 Classification setup

Three models were evaluated in a classification setting:

- Original Bayesian Network (expert-defined structure)
- Learned Bayesian Network (best-performing model: constraint-based, 500 samples)
- Naive Bayes classifier (baseline model)

The class variable was *PhishingEmail*. The selected feature variables were *SenderReputation*, *EmailGrammar*, *ContainsLinks*, and *UrgencyLanguage*, as these represent observable characteristics available when analyzing an incoming email.

Performance was evaluated using ROC curves and Area Under the Curve (AUC) scores.

2.3 Results

2.3.1 Structure learning results

The results show that structure recovery improves significantly with increasing sample size. The constraint-based approach consistently produced more accurate network structures than the search-and-score based method. With 500 samples (as well as the 1000 samples variant), the learned network closely resembled the original structure.

2.3.2 Classification results

The classification performance of the three models is summarized in Table 2 and visualized in Figure 2.

The Naive Bayes classifier achieved the highest classification performance with an AUC of 0.967, despite its simplifying independence assumptions. This indicates that, for this specific problem, the selected phishing indicators provide strong predictive information individually. The learned Bayesian Network achieved a very similar performance to the original expert-defined network, with AUC values of 0.847 and 0.845 respectively. This demonstrates that the structure learning process was able to recover a model that captures the relevant dependencies from the generated data while maintaining predictive performance close to the original network.

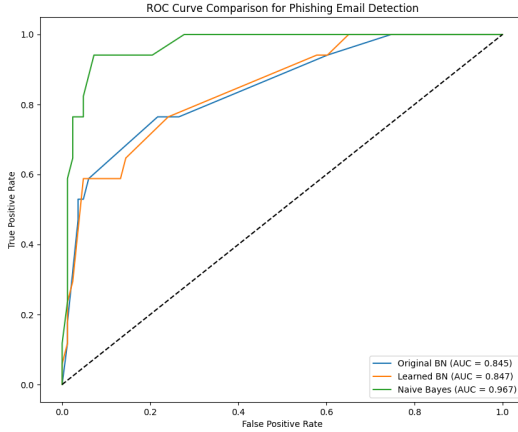


Figure 2: ROC Curves for the models

Model	AUC
Original Bayesian Network	0.845
Learned Bayesian Network	0.847
Naive Bayes classifier	0.967

Table 2: AUC scores for the models

2.4 Conclusion

This experiment demonstrates that Bayesian Network structure learning is influenced by both the available sample size and the selected learning strategy. The learned Bayesian Network achieved a classification performance comparable to the original expert-defined model, indicating that the learning algorithm was able to recover a useful representation of the underlying phishing process from synthetic data.

The Naive Bayes classifier achieved the highest AUC score (0.967), showing that a simpler model can perform very well when the selected features are strong predictors of the class variable.

However, this improved predictive performance comes at the cost of reduced interpretability, as Naive Bayes assumes conditional independence between features. The expert-defined Bayesian Network remains valuable because it explicitly represents causal relationships and provides insight into how different phishing indicators contribute to the final risk assessment.

References

- [AHNK21] Zainab Alkhalil, Chaminda Hewage, Liqaa Nawaf, and Imtiaz Khan. Phishing attacks: A recent comprehensive study and a new anatomy. *Frontiers in Computer Science*, 3:563060, March 2021.
- [JLN25] Raja Jabir, John Le, and Chau Nguyen. Phishing attacks in the age of generative artificial intelligence: A systematic review of human factors. *AI*, 6(8):174, 2025.
- [Nai15] Rennie Naidoo. Analysing urgency and trust cues exploited in phishing scam designs. 01 2015.
- [PBP⁺16] Kathryn Parsons, Marcus Butavicius, Malcolm Pattinson, Dragana Calic, Agata McCormac, and Cate Jerram. Do users focus on the correct cues to differentiate between phishing and genuine emails? 2016.
- [Ros25] Rossy Halimatun Rosyidah. Semantic urgency and illusion of authority in phishing emails: A corpus-based analysis. *Journal of Linguistics, Literature, and Language Teaching (JLLLT)*, 5(1):10–23, August 2025.
- [ZBDR⁺24] Sijie Zhuo, Robert Biddle, Jared Daniel Recomendable, Giovanni Russello, and Danielle Lottridge. Eyes on the phish(er): Towards understanding users’ email processing pattern and mental models in phishing detection. In *Proceedings of the 2024 European Symposium on Usable Security*, page 15–29, Karlstad Sweden, 2024. ACM.